JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai

CT 1 Examination

Course: B.Tech. (CSE)/(AI)
Subject: Formal Languages and Automata
Year/Semester: IV Sem (2nd Year)
Date:06/02/2025

Subject Code: CS4T003
Max Marks: 20
Duration: 1 Hr.

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Course Outcome: Students will be able to:
CO1: Define the mathematical principles behind theoretical computer science.
CO2: Differentiate among different types of automata like finite automata, push down automata, linear bounded automata and turing machine.
CO3: Correlate the different types of automata to real world applications.
CO4: Choose and design appropriate automata for the different requirements outlined by theoretical computer science.
CO5: Identify the different computational problems and their associated complexity.

SECTION A

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Draw the Finite Automata model.	L1/CO1	1
Q.2	State the five tuples of Finite Automata.	L1/CO1	1
Q.3	List out the applications of Finite automata.	L2/CO1	1
Q.4	Define Moore and Mealy machine	L2/CO1	1

SECTION B

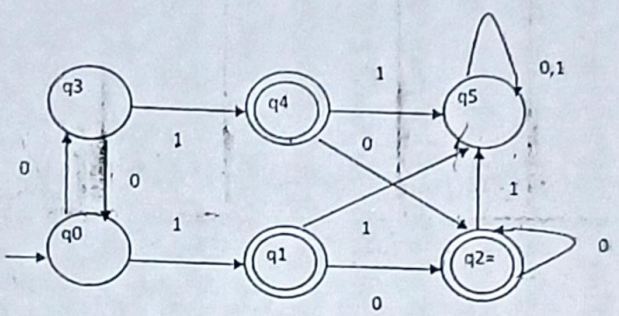
1. All Questions are compulsory.

Q.5	Design a DFA with $\sum = \{0, 1\}$, which accepts all string ending with two consecutive 0's.	L2/CO4	2
Q.6	State the difference between a^* and a^+.	L2/CO2	2
Q.7	How many number of states are required to design the automata for the strings containing 'aba' as a substring.	L3/CO1	2
Q.8	Design a Moore machine to determine residue mod-3 for each binary string as integer.	L4/CO2	2

SECTION C

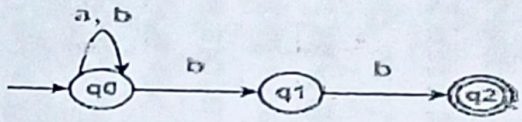
1. **Solve any two questions.**

Q.9 Minimize the following DFA.



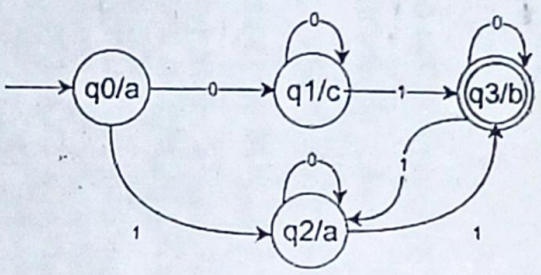
L2/CO3 4

Q.10 Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA).



L2/CO2 4

Q.11 Convert following Moore machine into its equivalent Mealy machine.



L1/CO2 4